Summary in Graph

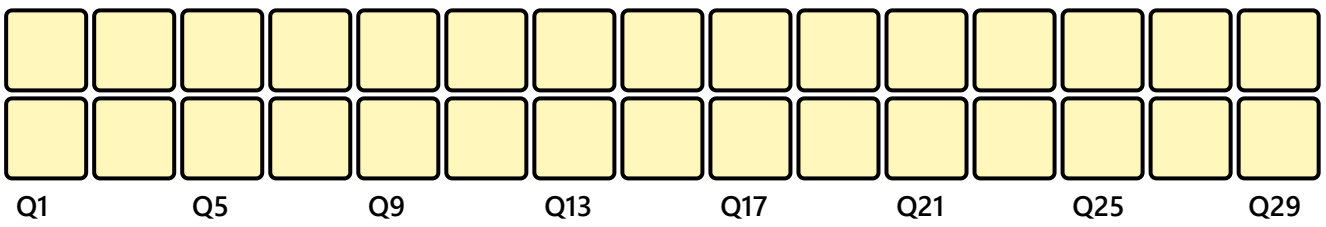
Exam Summary (GATE Overflow CS Test Series 2026 | Probability | Shallow Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (2) out of 15 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

A company has 7 departments and 6 tasks to assign in a particular day. Assume that the tasks are assigned randomly to departments, with all possibilities for which task goes to which department equally likely. What is the probability percentage that some department gets more than 1 task?

A. more than 85%

B. more than 95%

C. less than 85%

D. less than 95% and more than 90%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Probability

Consider the following series of numbers:

$$2, 5, 17, 9, 17, 7, 11$$

If a denotes the mean, b denotes the mode, and c denotes the median of these numbers, what is the value of $3c + b - 2a$?

Your Answer: | Correct Answer: 24.57:24.58 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Probability

If each value in a dataset of 8 values is multiplied by 7, the percentage change in the standard deviation will be _____.

Your Answer: | Correct Answer: 600:600 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following statements about the Poisson distribution is/are correct? (Mark all the appropriate choices)

A. The cumulative probability for a Poisson random variable X with parameter λ for at most k values is given by $P(X \leq k) = \sum_{i=0}^{k} \frac{\lambda^i e^{-\lambda}}{i!}$.
B. For a Poisson random variable with parameter λ, if λ is a non-integer, the probabilities for all integer values of k can still be computed using $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$.
C. If X is a Poisson random variable with parameter λ, then the distribution of X approaches a normal distribution as λ increases.
D. A Poisson random variable can take on negative values if the rate λ is negative.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following statements with respect to the random variable X following binomial distribution is/are correct?

Here, p and n represent the probability of success, and the number of trials, respectively.
(Mark all the appropriate choices)

A. The sum of two independent binomial random variables with different p values is also a binomial random variable
B. The expected value of X is always greater than its variance

C. The binomial distribution can only be used when the trials are independent.
D. For a binomial random variable X, as n increases and p stays fixed, the distribution of X approaches a normal distribution (under certain conditions).

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Three dice are rolled. The probability of getting at least one die showing a number greater than 4 is ____. (Rounded to two decimal points)

Your Answer: | Correct Answer: 0.70:0.70 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Probability

A biased coin is flipped 7 times. The probability of getting heads on any flip is $\frac{2}{3}$ and tails is $\frac{1}{3}$. If the probability of getting either **all heads** or **all tails** is Y, then $1000Y =$ ___ (Rounded to nearest integer).

Your Answer: | Correct Answer: 59.0:59.0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Probability

A collection contains 8 slips of paper, each marked with a letter from the word $\mathrm{GATE2025}$. What is the probability that if the letters are drawn in sequence with replacement, they will form the exact word $\mathrm{GATE2025}$? If this probability is denoted as Z, $10000000Z =$ ____ (rounded to two decimal places).

Your Answer: | Correct Answer: 2.38:2.39 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A biased coin is flipped six times, and the probability of getting a head on a single flip is 0.3. What is the probability of getting exactly 4 heads in these 6 flips?

A. $\frac{15}{28}$
B. $\binom{6}{4}$
C. 0.0595
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Probability

The expected value of the number of tails when flipping three fair coins is ____.

Your Answer: | Correct Answer: 1.5:1.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Assume there are at least n different people in a room. If the probability that at least two of them have the same favorite color is greater than 15%, what is the least possible value of n given that there are 10 different colors available?

A. 2
B. 5
C. 3
D. 4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

If you randomly select four points along the circumference of a unit circle, creating four distinct arcs, what is the expected length of the arc that includes the point $(0, 1)$?

A. $\frac{2\pi}{5}$

B. $\frac{3\pi}{2}$

C. π

D. $\frac{4\pi}{5}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A baker keeps mixing batches of dough, intending to continue until he has produced at least one loaf of bread and at most one batch of cookies. If the probability of each batch being a loaf of bread or cookies is equal, and he can keep mixing indefinitely, what is the expected number of batches he will mix?

A. 3
B. 2
C. 6
D. 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | Probability

A ribbon measuring 5 meters is cut into two segments by selecting a point uniformly at random along its length. What is the expected length of the longest segment (in meters)?

Your Answer: | Correct Answer: 3.75:3.75 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Numerical Type | Award: 2 | Penalty: 0 | Probability

Suppose that on a given day, the number of emails received by a customer service desk follows a Poisson distribution with a mean of 3.2 emails per day. The probability that the customer service desk will receive at most 5 emails in a day, in percentage, is _____ (up to one decimal point).

Your Answer: | Correct Answer: 89.5:89.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose the probability that a light bulb will fail during any one hour period is 0.05. The probability that a given light bulb will work for 4 consecutive hours is ____?

A. 0.90
B. 0.82
C. 0.81
D. 0.80

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Assume that a random variable X is normally distributed with mean $\mu = 10$ and variance $\sigma^2 = 25$. The probability that X is between 8 and 12 is ___. (Rounded to 2 decimal points)

$$\left(\phi(0.4) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0.4} e^{-x^2/2} dx = 0.6554\right)$$

Your Answer: | Correct Answer: 0.31:0.31 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Consider that a computer hard drive has an average lifespan of 500 days. Assuming exponential distribution, what is the absolute value of the difference in probability that it lasts more than 600 days and the probability that it lasts less than 400 days? Provide your answer rounded to two decimal points.

Your Answer: | Correct Answer: 0.25:0.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 2 | Penalty: 0 | Probability

In a toy factory, types of toys A, B, C, and D account for 40%, 25%, 20%, and 15% of the total production, respectively. Of their output, 3%, 1%, 4%, and 2% respectively are found to be defective. If a toy is randomly selected from the production and is found to be defective, what is the probability that it was manufactured as type D? (Answer Rounded to 2 decimal points)

Your Answer: | Correct Answer: 0.11:0.12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 2 | Penalty: 0 | Probability

A researcher collected two sets of data representing the growth rates of two different plant species over a month: 2.5%, 3.0%, 4.5%, and 5.0% for species A, and 2.0%, 2.8%, 4.0%, and 4.2% for species B. Let the geometric mean of the growth rates for species A be g_1 and for species B be g_2. Instead of calculating the

geometric mean uplift g_u as g_1/g_2, the researcher computed the geometric mean of the ratios of the two species' growth rates: $2.5/2.0, 3.0/2.8, 4.5/4.0$, and $5.0/4.2$. The percentage deviation of the value obtained from g_u is ____ (Rounded to one decimal point).

Your Answer: | Correct Answer: 0:0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a series of 3 distinct values $a < b < c$ where a, b, and c are in arithmetic progression, the values of a and c are both increased by 100%. Which of the following statements is/are TRUE? (Mark all the appropriate choices) A. The arithmetic mean increases by 66.67% B. The geometric mean increases by 41.4% C. The arithmetic mean increases by 100% D. The geometric mean increases by 50%

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Probability

A factory has two production lines: Line 1 and Line 2. Line 1 produces 60% of the factory's total widgets, while Line 2 produces 40%. Line 1 has a defect rate of 5%, whereas Line 2 has a defect rate of 10 A quality inspector randomly selects a widget from the factory's total production. After testing, the inspector finds that the selected widget is defective. Given that the widget is defective, what is the probability that it was produced by Line 1?

Your Answer: | Correct Answer: 0.42:0.43 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Select Type | Award: 2 | Penalty: 0 | Probability

In a bag of fruit, there are 16 pieces in total, consisting of apples and oranges. The probability that two randomly selected pieces of fruit are the same type is equal to the probability that they are of different types. Which of the following are the correct combinations for the number of apples (a) and grapes (g) in the bag? (Mark all the appropriate choices)

A. $a = 4, g = 12$
B. $a = 10, g = 6$
C. $a = 6, g = 10$
D. $a = 12, g = 4$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A university has two departments: the Science Department and the Arts Department. The Science Department enrolls 70% of the total students, while the Arts Department enrolls 30%. The average pass rate for the Science Department is 90%, whereas the Arts Department has a pass rate of 80%.
You randomly select a student who has failed their course. What is the probability that this student is from the Arts Department?

A. $\frac{2}{5}$
B. $\frac{3}{7}$
C. $\frac{6}{13}$

D. $\frac{1}{2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Probability

A job recruitment agency has two types of job listings: $Remote$ and $On-Site$. Suppose that 60% of all job listings are Remote. In 15% of the Remote job listings, the phrase "flexible hours" is used, whereas this phrase is only used in 5% of On-Site job listings. A new job listing has just been posted, which mentions "flexible hours". What is the probability that it is a Remote listing? (Answer up to 2 decimal places)

Your Answer: | Correct Answer: 0.81:0.82 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Two bags of weights are in a room. Bag A contains 4 light weights (each weighing 1 kg) and 6 heavy weights (each weighing 5 kg). Bag B contains 7 light weights (1 kg each) and 3 heavy weights (5 kg each). One of the bags is randomly chosen, and two weights are randomly picked from this bag without replacement. Both weights are light (1 kg each). What is the probability that the chosen bag is Bag B, given that both weights are light?

A. $\frac{5}{9}$
B. $\frac{6}{9}$
C. $\frac{7}{9}$
D. $\frac{8}{9}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Probability

An unbiased die is rolled repeatedly, and the outcome is recorded. What is the expected number of rolls to get a specific number, say 4, twice in a row?

Your Answer: | Correct Answer: 42:42 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. Dividing each number in a series by a constant decreases its arithmetic mean by the same constant.
B. Subtracting a constant from each number in a series decreases its geometric mean by the same constant.
C. Adding a constant to each number in a series does not change its standard deviation.
D. Multiplying each number in a series by a negative constant reverses the order of the data points.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | Probability

A bakery has 15 different types of pastries, consisting of 6 fruit pastries and 9 cream pastries. Assuming that all selections are equally likely, what is the probability that the 5 pastries chosen for the display case are all fruit pastries? (Answer Up to 3 decimal places)

Your Answer: | Correct Answer: 0.001:0.002 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A fair coin is flipped repeatedly until it lands on heads. If the total number of flips n is odd, what is the probability that $n = 5$?

A. $\frac{3}{64}$
B. $\frac{5}{32}$
C. $\frac{1}{2^5}$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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